PyGame Pong Tutorial - Part 4

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Time required: 30 minutes

Yes, we are finally going to make something move. By changing the (x, y) values each time through the game loop, we animate our ball.

Bouncing Ball

1. Save **pong_3.py** as **pong_4.py**
2. Add the following code to check for collisions between the ball and the walls.

```
# -------------------- CHECK COLLISION ----------------------------------#
    def check_collision(self):
        """Check for collisions with all 4 walls"""
        # Check for collision with left or right wall
        if self.ball.left < 0 or self.ball.right >= config.WIDTH:

            # Reverse y direction multiply by -1
            self.ball_speed_x = self.ball_speed_x * -1

        # Check for collision with top or bottom wall
        if self.ball.top < 0 or self.ball.bottom >= config.HEIGHT:

            # Reverse y direction multiply by -1
            self.ball_speed_y = self.ball_speed_y * -1
```

Draw Net

Let's draw a net.

```python
# ------------------------ DRAW NET ---------------------------------------#
    def draw_net(self):
        # Define the width of the net lines
        net_width = 2

        # Loop through the height of the game screen
        # with a step of 20 pixels
        for i in range(0, config.HEIGHT, 20):

            # Draw a rectangle representing a part of the net
            pygame.draw.rect(
                self.surface,   # Surface to draw on
                config.WHITE,   # Color of the rectangle (white)
                (               # Rectangle coordinates and size
                    # X-coordinate of the left corner of the rectangle
                    config.WIDTH // 2 - net_width // 2,
                    i,  # Y-coordinate of top corner of rectangle
                    net_width,  # Width of the rectangle
                    10          # Height of the rectangle
                )
            )
```

Call the check_collision and draw_net method each time through the game loop.

```python
# --------------------------- GAME LOOP ---------------------------------#
    def game_loop(self):
        """Infinite Game Loop"""
        while True:
            self.check_events()
            self.check_collision()

            # --------------- DRAW ON BACKBUFFER -------------------------#
            # Draw everything on the backbuffer first
            # Fill the display surface with black
            self.surface.fill(config.BLACK)

            self.draw_net()
```

Set Random Ball Direction

When the ball is initially drawn, let's randomize which direction it goes.

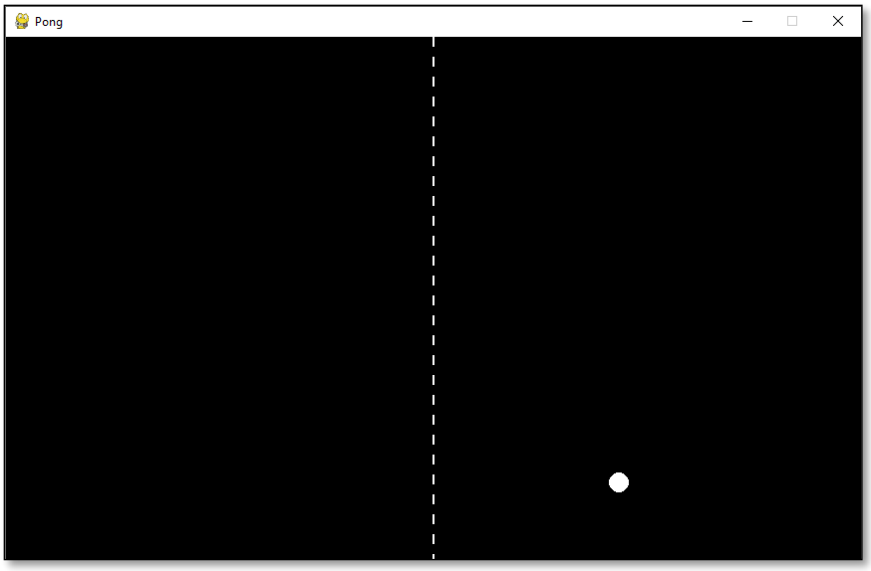
Add the following method.

```python
# ------------------------- SET BALL DIRECTION ------------------------------#
    def set_ball_direction(self):
        """Set initial ball direction along the x and y axis"""
        # Randomly determine the initial direction of the ball
        # along the x-axis (left or right)
        ball_direction_x = randint(0, 1)

        # If the randomly chosen direction is 0 (left),
        # set the horizontal speed of the ball to move to the right
        if ball_direction_x == 0:
            self.ball_speed_x = 3

        # If the randomly chosen direction is 1 (right),
        # set the horizontal speed of the ball to move to the left
        else:
            self.ball_speed_x = -3

        # Randomly determine the initial direction of the ball
        # along the y-axis (up or down)
        ball_direction_y = randint(0, 1)

        # If the randomly chosen direction is 0 (up),
        # set the vertical speed of the ball to move downwards
        if ball_direction_y == 0:
            self.ball_speed_y = 3

        # If the randomly chosen direction is 1 (down),
        # set the vertical speed of the ball to move upwards
        else:
            self.ball_speed_y = -3
```

Make these changes to the init method.

```python
"""
    Name: pong_4.py
    Author:
    Date:
    Purpose: Bouncing Ball
"""
# pip install pygame-ce
# Import pygame library
import pygame
# Import sys.exit to cleanly exit program
from sys import exit
from random import randint
import config


class Pong:

    def __init__(self):
        # Initialize pygame library
        pygame.init()

        # Set screen width and height as a tuple
        self.surface = pygame.display.set_mode(
            (config.WIDTH, config.HEIGHT)
        )

        # Set window caption
        pygame.display.set_caption("Pong")

        # Setup a computer clock object to keep the
        # game running at a constant speed regardless of computer speed
        self.clock = pygame.time.Clock()

        # Create the ball Rectangle object
        self.ball = pygame.Rect(
            config.WIDTH // 2 - config.BALL_RADIUS,     # Set x-coordinate
            config.HEIGHT // 2 - config.BALL_RADIUS,    # Set y-coordinate
            config.BALL_RADIUS,     # Set width of ball
            config.BALL_RADIUS      # Set height of ball
        )

        self.set_ball_direction()
```

Example run:



The ball bounces all over the place, but stays in the playing field.

Paddles are next.

Assignment Submission

Zip up the program files folder and submit in Blackboard.